Efficiency first, then solar: sequencing investments in a Mexican footwear SME

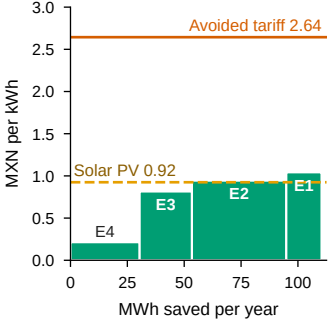
Data a typical SME already has

- 28 monthly bills
- Production records
- Walk-through audit
- Official tariffs, emission factor, fuels
- NASA POWER weather

Bills reproduced within 2.2%

1 Cut waste

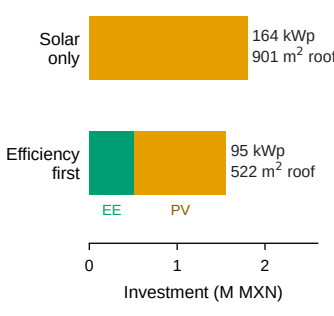
Cost of saved electricity



−23% grid electricity for 0.52 M MXN

2 Then size solar

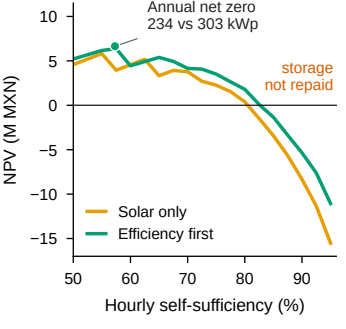
Same 54% cut in grid purchases



−13% capital, −42% roof, higher NPV in 86% of runs

3 Net-zero electricity

Least-cost design per hourly self-sufficiency



Annual net zero pays; deep hourly autonomy does not

Security and resilience

- Peak demand 157 → 124 kW with efficiency first
- If PV loses demand credit: solar NPV −69%, efficiency unaffected
- 111 tCO_2 and 850 GJ of imported gas avoided per year